Written Exam in Algorithms and Data Structures Spring 2024

This document contains ***9 pages***, including the front matter.

Response can be written in danish or english.

The exam set contains five questions which can be solved independently of each other.

Within each question there are multiple activities, with increasing order of difficulty.

Remember:

- Move on to other questions and activities if you are struggling with a particular activity. Come back to it later when you have time.
- For algorithms that are solving optimization problems, an algorithm that returns an approximate solution is better than no algorithm.

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Question 1 (15%): Algorithm Analysis

Consider the following algorithm:

```
procedure MyAlgorithm(A : list of integers)
    n := length(A)
    assert n>1
    swapped := true
    while swapped do
        swapped := false
        for i := 1 to n-1 inclusive do
            if A[i-1] > A[i] then
                swap(A[i-1], A[i])
                swapped := true
            end if
            if A[n-i-1] > A[n-i] then
                swap(A[n-i-1], A[n-i])
                swapped := true
            end if
        end for
    end while
end procedure
```

Provide an answer to each the following:

1. Detail the algorithm steps for given `A = [2, 4, 3, 7, 1]`.
2. Explain what the goal of the algorithm is. In other words, what does it do to `A`?
3. Write down the mathematical expression describing the maximum number of steps taken by the algorithm for a given `A` of size `n`.
4. What would `A` have to look like to trigger the maximum number of steps case?
5. What's the big-O worst case time complexity for this algorithm?
6. Can you think of ways to optimize the algorithm? If so, how would you modify the pseudo code to perform those optimizations? Would the big-O worst case time complexity change with these optimizations?

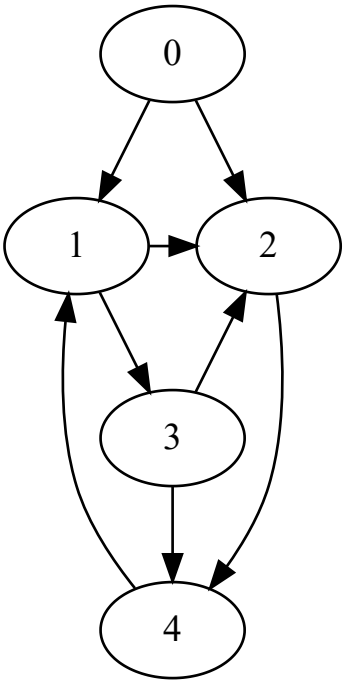
Question 2 (20%): Recursion

Consider the following c++ code for a recursive algorithm that detects cycles in a given connected directed graph.

```
// Returns true if the graph contains a cycle, else false
bool isCyclic(std::vector<std::vector<int>>& graph,
              const int N)
{
    // Mark all the vertices as not visited
    // and not part of recursion stack
    std::vector<bool> visited(N, false);
    std::vector<bool> recStack(N, false);

    for (int v = 0; v < N; v++) {
        if (!visited[v]
            && isCyclicRec(v, graph, visited, recStack, N)) {
            return true;
        }
    }

    return false;
}
```

```
bool isCyclicRec(int v,
    std::vector<std::vector<int>>& graph,
    std::vector<bool>& visited,
    std::vector<bool>& recStack,
    const int N)
{
    if (visited[v] == false) {
        visited[v] = true;
        recStack[v] = true;
    }

    for (int i = 0; i < N; i++) {
        if (graph[v][i] != 0) {
            if (recStack[i]) {
                return true;
            }
            if (!visited[i]
                && isCyclicRec(i, graph, visited, recStack, N)) {
                return true;
            }
        }
    }
    // Remove the vertex from recursion stack
    recStack[v] = false;
    return false;
}
```

1. Draw/Explain the different steps the algorithm takes for the following graph. What is the returned result of the algorithm? Which node does it visit first? Is node 3 ever visited?



2. Describe which is(are) the base case(s).

3. Describe which is(are) the recursive case(s).

4. Do you need to modify the algorithm to handle disconnected graphs? Write the adjusted code or justify your answer.

~~Question 3 (15%): String Searching~~

Given the following string and search pattern, use Knuth-Morris-Pratt to solve the following questions.

1. Calculate the prefix, *pi*, for the search pattern, *P*, given in the following:

i		**0**	**1**	**2**	**3**
P		a	b	a	a
pi	-1	?	?	?	?

2. Show the steps involved to search the following string, *S*. Also include how the prefix is used to find the pattern position:

i	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
S	a	b	a	b	c	a	b	a	a

Question 4 (15%): Binary tree-based data structure

Given an unsorted list of `N` social network users and their influencial level, which indicate their number of followers:

Username	Influencial level
Alice	2
Bob	8
Charlie	1
David	10
Eve	9
Frank	6

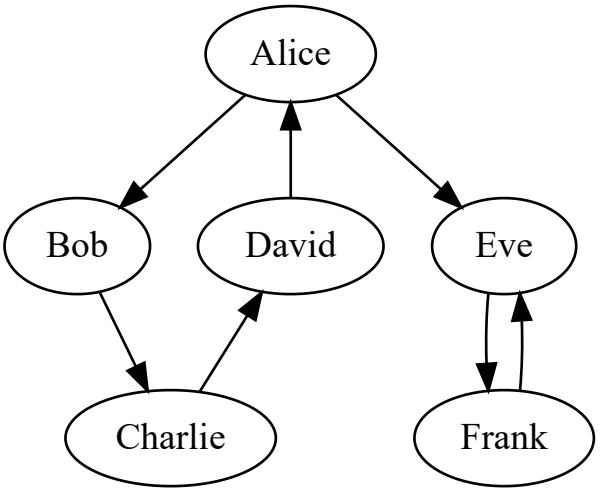
Say we want to create a function to provide a list of the `k` (`k=[1..N]`) most influencial users:

1. Which binary tree-based data structure (used in class) **is the best** for generating such a top-k list? Argument why you choose that particular structure.
2. Say we have the list above with 6 users. Illustrate the steps of your chosen algorithm to build a top-three list.
3. What is the worst-case time complexity of the algorithm?

Question 5 (35%): Graph

In a social network like X/Linked/Facebook/Instagram, users can follow other users. Consider a simple social network where when user A makes a post, then all A's followers will see the post. If user B is a follower of A, then B can also repost any post from A, allowing B's followers to see the post from A. The social network software keeps track of who follows whom.

Here's an example graph representing the social network:



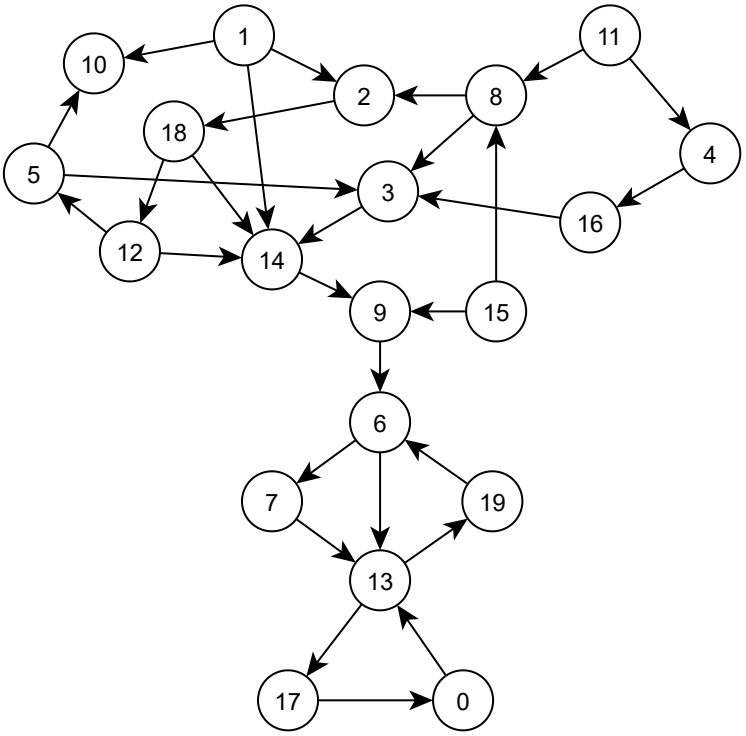
Questions:

1. Each node is the unique user name. Which data structure would you recommend using to store the nodes of the graph? Justify your answer.
2. What is the meaning of a cycle in this graph? What can happen if a post is made by one of the users belonging to a cycle?
3. Draw the adjacency matrix of the above given graph. Assume that the index corresponding to each username is as follows:

Username	Index
Alice	0
Bob	1
Charlie	2
David	3
Eve	4
Frank	5

4. Suppose you are a company who wants to advertise a given product in the social network. Which user would you approach in the following graph,

assuming that users are not allowed to repost. To simplify, the usernames have been replaced by integer numbers. Argue for your answer, and write the pseudo code for the algorithm that computes the answer for any given graph.



5. Suppose you are a company who wants to advertise a given product in the social network. Which user would you approach in the graph of the previous activity, **assuming that a user reposts 50% of the posts the first time she/he sees it**. To simplify, the usernames have been replaced by integer numbers. Argue for your answer, and write the pseudo code for the algorithm that computes the answer for any given graph.